TCR–Epitope Binding Prediction

Next Steps & Roadmap

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1. Where We Are

A classical ML baseline pipeline is complete. Random Forest with ESM2 embeddings achieves **AUC 0.738**. UMAP analysis confirms that the joint TCR+Epitope embedding space has biologically meaningful structure (pathogen-specific clusters, MHC class separation). The next phase focuses on improving negative generation, building a proper evaluation framework, and moving to graph-based models.

2. Roadmap

#	Step	Description	Status
1	**Improve negatives**	Epitope-aware mismatched pairing: for each positive pair, the negative TCR comes from a different specificity group, reducing false negative contamination.	Done
2	**Epitope-split evaluation**	Train/test split via `GroupShuffleSplit` by epitope so no epitope appears in both splits. Tests true generalisation to unseen epitopes.	Done
3	**Embedding analysis**	UMAP projections of TCR and epitope embeddings separately. Silhouette score and intra/inter distance analysis quantify clustering by epitope.	Done
4	**Larger ESM2 model**	Replace `esm2_t6_8M` with `esm2_t12_35M` or `esm2_t30_150M` for richer representations. Expected AUC improvement of 2–5 points.	
5	**Better negatives v2**	Generate negatives by pairing a TCR with an epitope from a different V-gene cluster. Biologically motivated and harder for the model to exploit spuriously.	
6	**Knowledge graph integration**	Use the iggytop Neo4j graph to add graph-level scalar features: TCR promiscuity score, epitope immunodominance score, MHC restriction pathway.	
7	**GNN model**	Bipartite graph: TCR nodes + epitope nodes + binding edges. GraphSAGE or GAT for link prediction. Compare AUC against RF baseline.	Planned
8	**Cross-dataset validation**	Evaluate the final model on an external holdout (IEDB or McPAS-TCR) to test generalisation beyond iggytop.	Planned

3. Priority for Next Session

The current pipeline uses the smallest ESM2 variant (8M parameters). Upgrading requires a single config change and a cluster re-run. This is expected to be the highest-impact single improvement.

- **Code:** set `ESM_MODEL = "facebook/esm2_t12_35M_UR50D"` in `tcr_epitope_minimal.py`
- **Cluster:** re-run `submit_minimal.slurm` on `sacramento` node (A100 40 GB)
- **Evaluation:** compare AUC vs current baseline of 0.738

The iggytop database already exposes a Neo4j knowledge graph. The goal is to extract scalar graph features and add them to the ESM2 feature matrix.

- Run `create_knowledge_graph.py` to populate the local Neo4j instance
- Query: for each TCR node, how many distinct epitopes is it connected to? (*promiscuity score*)
- Query: for each epitope node, how many TCRs bind it? (*immunodominance score*)
- Append these scalar features to X alongside ESM2 embeddings and re-train RF

Based on the embedding analysis, the recommended GNN design is:

- **Graph type:** bipartite — TCR nodes (features = ESM2 TCR emb.) and epitope nodes (features = ESM2 epitope emb.)
- **Edges:** known binding pairs from the training set (positive examples)
- **Task:** link prediction — does an edge exist between TCR_i and epitope_j?
- **Architecture:** GraphSAGE (inductive; handles unseen nodes at inference) or GAT (attention over neighbours)
- **Library:** PyTorch Geometric (`torch_geometric`)

4. Expected Timeline

Week	Goal	Deliverable
Week 1	Larger ESM2 + re-evaluation	New AUC benchmark; updated results_summary.tex
Week 2	Knowledge graph feature extraction	Graph scalar features added to feature matrix; AUC comparison
Week 3	GNN prototype (GraphSAGE)	First GNN AUC vs RF comparison in MLflow
Week 4	GNN tuning + cross-validation	Final model evaluation; report draft

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